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Scientifically Automating Data Quality Decisions with AI Explainability Weights

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ABSTRACT
This paper introduces Explainability Driven Data Quality (EDDQ), a methodology that leverages machine learning explainability weights to automate most data quality management decisions for machine learning models. EDDQ automates decisions end-to-end throughout the data quality process including decisions for: data catalog creation, Key Data Element identification, lineage documentation, data change management, attribute validation, and data quality issue management.

INTRODUCTION

The starting point for every machine learning project is: do you have the right data? And is the data you have in the right condition? The project then moves to the next question: is your data correct down to the attribute level? And then, to the quality of those attributes. Once everyone agrees that the data passes this quality assessment, you can then identify how important each attribute is to your data quality analytics.

The data quality of the most import attributes, often termed Key Data Elements (KDEs) or Critical Data Elements (CDEs), is then prioritized. Key Data Elements (KDEs) are subject to more comprehensive data quality assessments, and any failures identified in these elements necessitate more substantial remediation measures than those associated with less critical attributes.

Adhering to the aforementioned DQ best practices has been shown to significantly increase the likelihood of project success, while neglecting them almost invariably leads to failure. Traditionally, manually implementing these practices - such as building a data catalog and identifying Key Data Elements (KDEs) - has been both time-consuming and prone to error. This paper introduces an automated approach to streamline these processes, along with downstream Data Quality Management (DQM) practices, for data pipelines that feed machine learning (ML) models.

We present a new DQ framework: "Explainability Driven Data Quality" (EDDQ). The framework leverages machine learning and explainable feature weighting to automate most data quality management decisions for machine learning models. We present the framework in four sections:

1. An overview of the traditional Data Quality Management process.
2. An explanation of how automating DQM best practices shifts some areas of responsibility from human experts to machine learning models. In our case, using ML modelling to determine which data attributes qualify as Key Data Elements (KDEs).[1] We describe a typical DQ assessment scenario as an example of how current DQ assessments are sequenced, and how automation will alter that sequence.
3. A practitioner-oriented guide to implementing explainability in data quality practices.

[1] A Key Data Element is a specific data field that is critical to business operations, decision-making, compliance, or risk management. KDEs are prioritized in data quality assessments because errors in KDEs can cause significant business impact, and they are often required for reporting in regulated industries.

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4. A conclusion that includes challenges and future directions in implementing the EDDQ methodology and DQ automation.

During model training, a machine learning (ML) system learns which input attributes influence its decisions and to what extent. The use of explainability algorithms to automatically capture this information enables automation of much of the manual effort traditionally required in Data Quality Management (DQM). Before delving into explainability and its automation implications, we present an overview of the existing manual DQM process.

THE EXISITING MANUAL DATA QUALITY MANAGEMENT PROCESS

In the current manual DQM state, the aforementioned DQM best practices are widely accepted, but not always effectively followed, or at all. One common reason is that the data scientists who conduct Data Quality (DQ) assessments are asked to execute a Proof of Concept (POC) with the available data in hand, in its current condition. If their ML assessment is judged successful, and adequate, success can often lead to accelerated deployment timelines. For example, a data scientist might create a model that is 75% correct in a one or two week period. Stakeholders might surmise that performance could be pushed to a higher target, e.g. 90%, with minimal additional effort. But 90% performance might be impossible to attain without acquiring better quality data.

Other factors may also intervene in the strict adherence to best practices. In an attempt to improve model performance, data scientists may try to excessively tune the model, leading to overfitting. The team may not recognize data situations that cannot be solved by tuning and cleaning. Resource constraints - time and money - may also be factors that lead data science teams to decide that it is best to work with the data they have, rather than escalating DQ issues to senior management.

This paper explores one such scenario in depth: a machine learning (ML) model is discovered to have been trained on input data where 10% of records reflect a data quality issue. We examine how a DQ team, relying primarily on human judgment, navigated the issue—and how an automated DQ framework might have offered earlier detection, clearer remediation pathways, and more accountable decision-making. From this case, we introduce a practical framework and decision tree designed for DQ professionals and Chief Data Officers (CDOs) to apply automated principles in real-world settings. We posit that in addition to greater use of automated tools, data governance policies should place increased emphasis on data science teams acquiring high-quality data in the first place, rather than repeatedly focusing on cleaning poor-data quality inputs. Our goal is to bridge policy with implementation, enabling more rigorous, transparent, and scalable data quality practices appropriate to the new drivers placed by AI on DQ practices.

Consider the following scenario, broken into two parts: the initial decision process, and the subsequent DQIM (data quality issue management) root cause analysis conducted afterwards.

A Hypothetical and Typical DQM Scenario

A company runs an automated DQ assessment on all the data that it ingests. The firm finds that 10% of the values for a specific data attribute are bad. [2] *Stakeholders discuss the issue and the impact on the machine learning model that reports it. Unfortunately, the stakeholders are unable to objectively assess the severity of inputting 10% bad data, leading them to propose actions based on qualitative evaluation and individual judgment. Risk averse stakeholders favor shutting down the model until they resolve the DQ issue. Other stakeholders favor introducing a human in the loop to review the ML model's results before taking action. One stakeholder says he suspects the issue is minor and they can operate as usual – "we can reverse a few bad recommendations made by the model with minimal impact." Others disagree. The team arrives at a decision only after escalating the matter to senior management and consulting with data*

[2] A data quality **attribute** is a specific characteristic used to assess the fitness of data for its intended purpose.

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scientists regarding the significance of Attribute X to the model's outcomes. The prolonged decision-making process results in financial losses for the company.

The above scenario, whether frequent or infrequent in your organization, presents the case for evaluating the increased use of automation in the DQ process. Had the DQ team had access to a data-driven automated tool that assessed severity and recommended options for addressing it, stakeholders could have resolved divergent views without further consultation with data scientists and escalation to management. And quite possibly, avoid financial losses that had already been incurred.

Identifying and Prioritizing KDEs

Traditionally, the identification of required attributes and selection of Key Data Elements (KDEs) has been a manual, labor-intensive task performed by data owners. KDEs are identified and prioritized based on their influence on model outputs and their overall impact on analytical or operational outcomes. The process typically involves data owners reviewing all available attributes, inputting them into machine learning models, and documenting them within a data catalog. The goal is to ensure that the data attributes used in the data quality management process (DQM), particularly the KDEs, are of consistently high quality.

From the attributes listed in the data catalog, data owners have relied on domain expertise and judgment to manually designate the attributes they consider most critical as Key Data Elements (KDEs). Best practice and BCBS 239 state that the data owner should then document the lineage of each attribute from the System of Record (SOR) that produced it through every data pipeline hop until it reaches the target consumer.[3] As a data consumer, the data owner should then create a data contract with the data producers defining the data distribution that the producer will share, including the producer's data quality guarantees, and the consumer's lens, the subset of distribution attributes that the consumer ML model uses.

Whenever an event affecting the data contract occurs, the data subject matter experts (SMEs) must manually decide what to do. For example, whenever a producer makes a data change event, the data contract stakeholders must manually determine the impact to the ML model and the risk level of the change. Similarly, whenever a KDE fails a data quality check, the stakeholders involved must decide how important the issue is and what to do about it.

BACK TO OUR HYPOTHETICAL SCENARIO: ROOT CAUSE ANALYSIS

We now return to our hypothetical scenario introduced in the previous section. We left off with the DQ team having to escalate the DQ issue to management. Team members could not agree on a course of action after discovering that 10% of a specific data attribute were bad. After consulting with management and deciding upon their operational approach, the team investigates the DQ issue and discovers that Attribute X is not in the data catalog because it is the only attribute pulled from a different source that was overlooked when creating the data catalog. Furthermore, Attribute X was never designated as a KDE because the team did not realize the attribute's importance to the model.

A root cause analysis of the DQ issue determined that multiple things went wrong:

- *Since Attribute X was not in the data catalog, the data producer was unaware that the model depended on Attribute X. Therefore, the producer's data change management process did not identify that changing Attribute X would impact the consuming model. What if we had a tool that could automate the attribute discovery for data catalogs and data contracts?*

[3] BCBS 239, officially titled "Principles for Effective Risk Data Aggregation and Risk Reporting," is a set of regulations issued by the Basel Committee on Banking Supervision (BCBS). These principles aim to enhance the risk management and reporting capabilities of global systemically important banks (G-SIBs).

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- *The firm's data contracts only require validating the quality of KDEs, so the validations didn't check Attribute X and the consuming model was not alerted to bad incoming data. What if we had a tool that could automatically review all attributes in the data catalog, and identify and designate the KDEs?*
- *The KDE lineage documentation did not include Attribute X so it took time to find and contact the attribute's data producer. The data producer used a different name for Attribute X in their system so it took another meeting and more investigation to map Attribute X to the data producer system and make the fix.*

THE NEW PARADIGM OF EXPLAINABILITY DRIVEN DATA QUALITY

As we have presented in this two-step scenario, automating steps in this traditionally manual sequence is both desirable and feasible. A key advancement enabling this shift is the use of explainability weights—machine-generated indicators that quantify the relative importance of data attributes, replacing manual weighting processes. By leveraging these weights, the DQM tasks of identifying relevant attributes for the data catalog, and determining which qualify as Key Data Elements (KDEs), can be shifted from human experts to machine learning models. Furthermore, the granularity of the automated weights can support a more refined decision making model, that in most circumstances will outperform the decisions that are possible with binary systems based on either KDEs, or non-KDEs.

Importantly, quantifying the influence of each KDE not only enables automated, impact-aware responses for data quality issue management, but also supports stronger model governance by providing transparent justifications for feature prioritization and downstream decisions. Yet what exactly are explainability weights, how reliable are they, and how complex is their generation and interpretation?

Explainability Weights and their Role in DQ Assessment

Explainability weights are quantitative values that represent the contribution or relative influence of individual input features to the output of a machine learning model. These weights enable interpretation of how and why a model makes specific decisions, supporting transparency and interpretability of the ML model output. Common methods for deriving explainability weights include SHAP values (SHapley Additive exPlanations) (Lundberg & Lee, 2017), LIME (Local Interpretable Model-agnostic Explanations) (Ribeiro et al., 2016), and integrated gradients (Sundararajan et al., 2017).[4] These approaches provide feature-level attributions by analyzing the model's behavior locally or globally, helping stakeholders identify key drivers behind predictions, facilitating debugging and compliance. For the purposes of the DQM automation case we are presenting, we focus on global explainability weights, which quantify the overall contribution of each attribute to the model's decision-making process as a proportion of total influence.

Generating global explainability weights is handled internally in code that loads the ML model, input data, and runs in our example, the SHAP algorithm. For a model estimating weights for three attributes, X, Y, and Z, the output would look something like this:

- Attribute X: 51.35%
- Attribute Y: 28.65%
- Attribute Z: 20.00%

The data reported shows that 51.35% of the model decision is based upon the value of Attribute X, 28.65% is based upon Attribute Y and 20.00% is based upon Attribute Z. Research on the SHAP frame-

[4] Lundberg, S. M. & Lee, S-I. (2017). A Unified Approach to Interpreting Model Predictions. In Proceedings of the 31st Conference on Neural Information Processing Systems (NeurIPS 2017**).** Ribeiro, M. T., Singh, S. & Guestrin, C. (2016)**.** *"Why Should I Trust You?": Explaining the Predictions of Any Classifier*. Presented at *KDD 2016*. Sundararajan, M., Taly, A. & Yan, Q. (2017). *Axiomatic Attribution for Deep Networks*. Proceedings of *ICML 2017*, 3319–3328.

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work has demonstrated that SHAP values in this class of algorithms satisfy key model principles like accuracy, consistency, and the handling of missing features.[5] These properties help establish tthe explanations are reliable and aligned with how the model actually makes decisions. In the context of DQ assessment, SHAP values enable three key outcomes: (1) identifying which data elements have the most impact on key decisions, (2) providing clear, model-agnostic values that help support model governance and transparency, and (3) supporting systematic, scalable detection of root causes behind data quality issues.

PUTTING IT ALL TOGETHER: FRAMEWORK FOR AUTOMATING DQM

The following section provides a practical, implementation-focused framework for automating key Data Quality Management (DQM) tasks by leveraging machine learning model artifacts, specifically the data catalog and Key Data Elements (KDEs). The goal of the framework is to outline a repeatable process - a decision sequence - for embedding DQ automation into high-value, externally reported Tier 1 business processes, aligned with established regulatory frameworks and industry standards such as BCBS 239, DAMA-DMBOK, GDPR, HIPAA, and SOX.

The framework identifies eight decision points specifying how to extract and operationalize metadata to support automated execution of foundational DQM tasks. The decision points in sequential order are:

1. Identify features and create a data catalog for the input dataset
2. Identify Key Data Elements (KDEs) in the data catalog
3. Document lineage for the KDEs
4. Create a data contract between data producers and data consumers
5. Perform Data Change Management (DCM)
6. Optimize DQ validation within a limited time frame
7. Document and take operational action in response to DQ issues
8. Assign severity levels to DQ issues as part of Data Quality Issue Management (DQIM)

SHAP output includes a list of all attributes used by a model, e.g., all attributes for the data catalog, and the weights representing the relative importance of each attribute to the model's decision making. Knowing the importance of each attribute can inform and automate our KDE designation process, our DQ issue response process, and all other DQM decision-making processes.

Multiple Lines of Defense

A typical Chief Data Officer (CDO) solution to the challenges in the aforementioned scenario is to implement second and third lines of defense to periodically, and manually, validate the correctness of the data catalog and KDEs for a given ML model. Additional lines of defense are helpful, especially from a defensive quality standpoint. Regulators appreciate firms that consider the challenges previously outlined and implement additional lines of defense to address them.[6] Unfortunately, the efficacy of the additional defenses can be questionable given their after-the-fact and manual nature.

An additional challenge for data officers is due to data delivery SLAs. The most common traditional data validation processes allow data consumers to use data as soon as it is prepared, often without waiting for comprehensive data quality checks to complete. Thus, data quality issues can potentially impact business processes or outcomes.

[5] Lundberg and Lee, Ibid.

[6] The "three lines of defense model" was developed by the Institute of Internal Auditors (IIA). First formally presented in a 2013 IIA position paper titled: *The Three Lines of Defense in Effective Risk Management and Control.* Revised and expanded in 2020 as *"The Three Lines Model."*

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Reactive data quality validation systems, e.g., compliance systems for Anti Money Laundering, Trade Surveillance, and Disclosure of Interest, can often replay the pipeline after correcting the data and regenerating their reports with minimal impact beyond the effort devoted to the replay. However, decision-making systems, e.g., check cashing, mortgage application, and client onboarding often struggle with reversing a decision made with bad data.

A PRACTITIONER'S GUIDE TO IMPLEMENTING EXPLAINABILITY IN DATA QUALITY PROCESSES

The above issues are the entry point for the promise of Explainability Driven Data Quality, a concept that leverages the global explainability weights output from an algorithm, e.g., SHAP, to drive data-driven, data quality decision making.[7]

A key innovation in this framework is to automatically extract the global explainability weights from ML models to determine which features are used (entered into data catalog), which features are weighted the highest (KDEs), and then to use the data catalog and KDEs to automate six subsequent key DQM processes. Including data cataloging, KDE identification, attribute lineage documentation, issue severity classification, contract enforcement, and data change management. All key steps in data quality issue management (DQIM). It is important to note that for traditional non-ML projects, the six subsequent DQM processes can be automated after manually creating the data catalog and tagging the KDEs.

Figure One presents the automated DQM steps in EDDQ, and the subsequent DQM steps in the sequence.

FIGURE ONE: AUTOMATING EDDQ

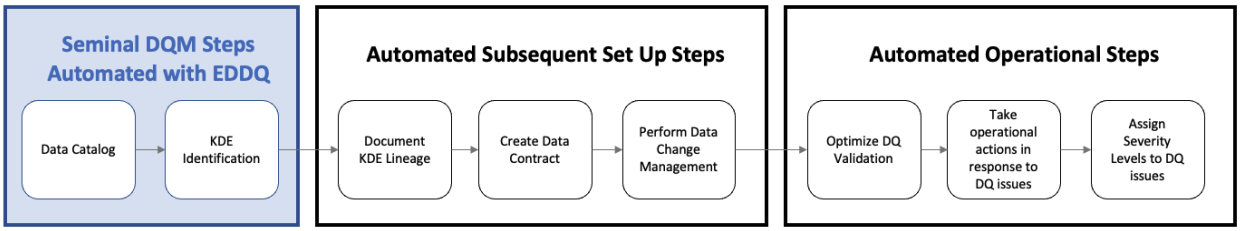

Table One provides a practitioner-level guide, organized as a table matching the target activity or process step to suggested automation strategies and procedures.

TABLE ONE: AUTOMATING EDDQ

Process	How to Automate
Global Explainability Weights	Automatically run a batch program to generate global explainability weights for all machine learning models with an algorithm such as SHAP. Extract the weights, sort them in descending order, and persist them.
Data Catalog Creation	Automatically review the explainability weights to identify all non-zero weights. Features with non-zero weights should be automatically added to the data catalog for that model.

[7] For readers interested in looking more closely at SHAP explainability models, and how these models compare with interpretable model designs and their relative design and usage considerations, see the Appendix for discussion.

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<table>
<tr><td>KDE Identification</td><td>Set an explainability weight threshold above which an attribute is considered a KDE. Automatically update attribute meta-data to designate KDEs as applicable.</td></tr>
<tr><td>Lineage Documentation</td><td>Document lineage via traditional (often manual) methods but prioritize the documentation in descending order of the automatically extracted and sorted feature explainability weights.</td></tr>
<tr><td>Data Contract Creation (Consumer Lens)</td><td>Automatically generate the consumer lens for data contracts from the data catalog (list of all attributes needed from the producer) and the KDEs (list of attributes requiring heavy DQ validation)

By defining a consumer lens, a consumer informs the producer of what the consumer actually consumes and expects so that the producer can notify consumers of just the issues that impact the consumer.</td></tr>
<tr><td>Quality Check Prioritization</td><td>Feed the global explainability weights into the perimeter data quality check process. In the commonplace event that there is not time to complete all data quality checks on all attributes for all data rows, ensure that checks receive attention in proportion to their importance. For example, if an attribute’s explainability weight accounts for 11% of a model’s decision, then spend 11% of the available time checking that attribute’s quality.</td></tr>
<tr><td>Data Change Management (DCM)</td><td>The DCM process should include checking the data contract to see if a consumer is affected by a data change. If the consumer is affected, then automatically set the change risk levels based upon the explainability weight, i.e., set higher risk levels for higher weights.

An automated system should also take into account whether multiple models use a feature and what the explainability weights are for each model in order to calculate the cumulative severity. Of course, the importance of the model is also an important factor.</td></tr>
<tr><td>Data Quality Issue Management (DQIM)</td><td>Automatically set issue severity levels and SLAs based upon the explainability weights of the feature in each model that uses an attribute. For example, a firm might automatically set severity levels such as:
<table>
<tr><th>Severity</th><th>SLA</th><th>Weight</th></tr>
<tr><td>Low</td><td>24 hours</td><td><= 2%</td></tr>
<tr><td>Medium</td><td>3 hours</td><td>2% > and <=5%</td></tr>
<tr><td>High</td><td>30 minutes</td><td>> 5%</td></tr>
</table>
As mentioned for DCM, an automated system should also take into account whether multiple models use a feature and the importance of those models.</td></tr>
</table>

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While the integration of EDDQ principles into data quality management offers a framework for governable, ML-ready data, it also introduces an initial uplift to build the components listed in the table as well as effort to maintain them. This effort is in addition to current efforts and thus must be acknowledged and addressed for the paradigm to be widely adopted and effectively implemented.

We consider the effort necessary to properly support investments that have already been made in machine learning, but not necessarily optimized. Managed properly, the effort required for EDDQ could be offset by the reduced operational effort post-automation and reduced data quality issues requiring clean up.

ISSUES AND CHALLENGES IN AUTOMATING DATA QUALITY PRACTICES

While the roots of data quality assessment can be traced back to the 1950s, data quality became a recognized field in IT and business operations in the late 1980s and 1990s.[8] Implementing greater automation in DQM involves important technical, cultural, and organizational challenges. Following are four key challenges stated by researchers, practitioners, and noted by speakers at recent DQ industry events.[9]

Organizational Mindsets, Ownership and Cultural Resistance

Historically, DQ has been a manual, stewardship-heavy discipline and sequence of practices. Data stewards and analysts view domain knowledge and judgment as critical, and in some respects, irreplaceable. In companies, DQ responsibilities may be distributed across different data teams, operations or business units, and each may have their own preferred methods and tools. Automation may also be seen as reducing headcount or shifting responsibilities away from DQ teams.

Technical and Architectural Challenges

What does "good data" look like? Without some level of standardized, domain specific rules or data dictionaries, there may be no consistent definition of "quality." Organizations using a patchwork of legacy systems may not be ready to support DQ automation. Lineage tracking, metadata management or schema registries are needed to implement automated checks that depend on context-aware model validation.

Lack of Data Governance – Technology Alignment

While many organizations have established DQ policies - such as compliance with international standards (e.g., ISO 8000) or protocols for bias detection and mitigation - these policies may remain high-level and declarative, rather than operationalized. Specifically, they are usually documented in human-readable formats (e.g., policy manuals, governance charters), which may not be readily interpretable by machines. Technically, effective data quality (DQ) automation requires that DQ tooling be embedded within, or tightly coupled to, production data and ML pipelines. However, DQ processes may be siloed or peripheral, operating independently from MLOps and DataOps workflows.

Change Management and the Challenge (and Cost) of Building Cross-Domain Expertise

While automation may be positioned as a fix to many data quality challenges, its success depends on the cross-domain expertise and experience that many DQ teams may lack. For example, the need to transition data stewards from manual data quality roles to DQ strategy, policy or rule authorship is frequently over-

[8] Companies including Trillium Software (founded in 1995) and DataFlux (also founded in 1995), and MIT's Total Data Quality Management Program, pioneered early software development, architectural principles, research and practical frameworks. See Richard Y. Wang, Diane M. Strong (1998). *A Product Perspective on Total Data Quality Management*. Communications of the ACM, Volume 41, Issue 2, February 1998. DOI: 10.1145/269012.269020

[9] Wang, R. Y., Madnick, S. E., & Pipino, L. L. (2006). Introduction to Information Quality. Cambridge, MA: MIT Information Quality Program. ISBN: 978-0-9790733-0-4. A key industry event is the annual Chief Data Officer Information Quality Symposium, CDOIQ.

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stated. Without significant change management, these manual roles lack the strategic exposure or systems-level experience required.

Addressing this and other experience gaps requires more than skills training. It requires new team structures, different cross-domain role experience - some of which has to be learned through practice, not "trained in" – and HR to take an investment posture in human capital development: building new team capabilities, different individual skillsets, and exposure to different roles and responsibilities that will be required to develop and manage automated systems.

FUTURE DIRECTIONS: TOWARD AUTOMATED DATA QUALITY DECISIONS

Despite these challenges, several emerging trends suggest a path forward – potentially resulting in the transformation of the data quality discipline itself. Forecasting the trajectory of data quality (DQ) automation requires careful consideration of current technological trends, organizational constraints, and evolving data infrastructure. While explainability techniques such as SHAP have advanced the interpretability of model-driven DQ assessments, they represent only one dimension of a broader movement toward intelligent, automated systems.[10] Emerging research in causal inference, unsupervised representation learning, and graph-based metadata modeling suggests the potential for systems that can not only detect anomalies but also infer their underlying causes with minimal supervision.[11] If adopted, these methods could enable more proactive DQ monitoring, where detection is tightly coupled with semantic understanding of data lineage, usage, and domain-specific constraints.

At the same time, trends in the operationalization of metadata—particularly through active metadata platforms and data observability frameworks - indicate a growing emphasis on context-aware automation.[12] This could support the development of semi-autonomous or autonomous remediation workflows, especially when combined with LLM-based interface layers that lower the barrier for rule authoring and diagnostics[13]. However, widespread adoption of such systems will depend on many factors, including the maturity of governance frameworks, trust in AI-driven interventions, and the integration of DQ processes within broader data lifecycle management practices such as DataOps and MLOps.[14] If these trends continue, DQ automation by 2030 may shift toward a more embedded, adaptive, and policy-driven paradigm—but such outcomes remain contingent on both technical progress and organizational willingness to invest in, and operationalize these capabilities.

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[10] Lundberg, S.M., & Lee, S.I. (2017). *A Unified Approach to Interpreting Model Predictions.* Advances in Neural Information Processing Systems, 30.

[11] Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press. Bengio, Y., Courville, A., & Vincent, P. (2013). *Representation Learning: A Review and New Perspectives*. IEEE Transactions on Pattern Analysis and Machine Intelligence, 35(8), 1798–1828. Zhao, Y., et al. (2020). *Automating Metadata Management with Graph Neural Networks*. ACM SIGMOD Record, 49(2), 27–32.

[12] Widad Elouataoui, Saida El Mendili, Youssef Gahi (2024). *Active Metadata and Machine learning based Framework for Enhancing Big Data Quality*. NISS '24: Proceedings of the 7th International Conference on Networking, Intelligent Systems and Security. 12 August 2024. Montecchiari, M., & Signorini, M. (2023). *Data Observability in Modern Data Infrastructure*. Data Engineering Journal, 2(1), 15–28.

[13] Bommasani, R., et al. (2021). *On the Opportunities and Risks of Foundation Models*. Stanford CRFM Report.

[14] Otto, B. et al. (2019). *Designing a Data Governance Framework*. Journal of Enterprise Information Management, 32(4), 550–574. Ehsan, U., & Riedl, M. (2020). *Human-Centered Explainable AI: Toward a Reflective Sociotechnical Approach.* ACM CHI Conference on Human Factors in Computing Systems.

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Future Research

The authors plan to continue exploring the role of AI explainability weights in advancing data quality, by introducing a framework, methodology, and practical guidelines for driving data quality decision-making in proportion to data attribute importance, as inferred from model behavior. This approach will seek to align data quality investment and validation intensity with the real-world influence of individual attributes on downstream AI outcomes. Important steps in this work will be to map attribute-level quality importance, and continued work on token-based and/or metadata-driven model architecture, among other topics.

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Disclosure

The analysis and views presented in this brief reflect the authors' independent work and are not intended to represent the views of any organization or individual affiliated with their employers, the SPARK AI Consortium, or affiliated organizations.

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APPENDIX: AUTOMATION, EXPLAINABILITY AND INTERPRETABILITY IN DQ DESIGN AND GOVERNANCE

The rapid proliferation of machine learning (ML) and artificial intelligence (AI) systems has introduced unique and massive data volume, velocity, and complexity demands into enterprise IT environments. With AI pipelines becoming increasingly data-dependent, the limitations of traditional, manual data quality (DQ) practices have become apparent. Conventional approaches - rooted in human review, static profiling, and periodic validation - are no longer adequate or best practice, for real-time, dynamic data environments where data drifts, schema shifts, and latent biases can undermine model integrity. A growing body of work now calls for automated, rule-based, and ML-driven DQ frameworks that can operate continuously and at scale.[15] The question is not whether, but where to automate responsibly.

Future automated data quality systems offer key advantages: enabling proactive anomaly detection through techniques like time-series modeling and statistical profiling, while ML-based monitors can learn data distributions and flag drift automatically.[16] Data contracts, expectations frameworks (e.g., *Great Expectations*), and constraint validation engines (*e.g., Deequ, TensorFlow Data Validation)* can provide reproducible enforcement of quality rules, reducing reliance on subjective judgment. Furthermore, metadata management, lineage tracking, and explainability tools (*e.g., OpenLineage, Amundsen)* can enhance auditability and root cause analysis, supporting reproducible data governance at scale. These systems are intended to enforce consistency across development and production environments, crucial for high-stakes AI deployments in regulated industries.

However, automation introduces its own set of risks. Static validation rules may fail to adapt to business context changes, while ML-driven DQ systems risk propagating the very biases they aim to detect if not properly governed.[17] Automated decisions made in opaque pipelines can be difficult to interpret or override, undermining trust among stakeholders. Additionally, the detection of semantic or contextual data issues—such as representational harm or sociotechnical bias—often requires cross-functional expertise that automation alone cannot replicate. These limitations point to the need for a hybrid model, combining automated enforcement with human-in-the-loop oversight and feedback loops.

More broadly, there is also a growing literature and challenge in “model explainability” versus “model interpretability” in design, execution and implementation in the first place. Work by Cynthia Rudin and others have focused on the imperative to design inherently interpretable model classes that are accurate and understandable to humans.[18] This typically means: no post hoc "explanations"; no black-box dependencies; and the principle of design by human-simulatable logic (e.g. so people can follow the decision process step-by-step). Rudin’s work cautions against unwitting over-automation of processes where the risk of error or harm is high, and “black box” design principles that are difficult to explain post-hoc to those using them. As pressure for automation increases, the questions posed by Rudin and others are what automation, where is it introduced, and what design principles are being used to keep humans in the loop.

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About the SPARK AI Consortium

The SPARK AI Consortium was founded in 2024 as a response to the rapidly evolving AI landscape and the growing need for interdisciplinary collaboration to address the complex challenges of AI implementation in business contexts.

Hosted at the San Diego Supercomputer Center (SDSC), SPARK AI brings together the technical expertise of Center AI researchers in advanced computation, data science, and integrative AI solutions with the expertise, experience, and practical insights of industry leaders to create a unique ecosystem for innovation and thought leadership.

The consortium is built around an eclectic mix of AI experts, senior data professionals, general managers, and academic researchers, united by a common goal: to ensure that AI technologies are developed and deployed in ways that create sustainable value for businesses while benefiting society as a whole.

An advisory board made up of corporate members, research faculty and industry experts guide the consortium in its strategy, operations and publication of research outputs.

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